

COLLEGE BUS ROUTE ANALYSIS REPORT

1. Problem Statement

The objective of this project is to analyse the usage of college bus routes and stops in order to identify inefficiencies and optimize transportation resources. The study focuses on understanding student distribution across different stops and suggesting improvements to enhance operational efficiency, reduce costs, and improve service quality.

2. Tools Used

- **Python** → Data processing and analysis
- **Pandas** → Data manipulation and aggregation
- **Matplotlib** → Data visualization
- **JupyterLab** → Development environment
- **CSV File** → Dataset storage

3. Data Description

The dataset contains the following columns:

- **student_id** → Unique identifier for each student
- **route_name** → Bus route assigned
- **stop_name** → Boarding location of the student
- **distance_km** → Distance from college
- **daily_usage** → Whether student uses bus daily
- **fee_paid** → Bus fee paid by student
- **students** → Number of students (used for analysis)

4. Visualization

The data was visualized using bar charts to compare student distribution across different stops.

5. Key Insights

- Student distribution varies more significantly at the **stop level** than at the route level
- Some stops have **higher student concentration**, indicating heavy usage
- Certain stops have **low student presence**, indicating inefficiency
- Uniform distribution across routes can hide inefficiencies, making stop-level analysis more effective

6. Route Improvement Suggestions (Most Important)

OVERLOADED STOPS

Stops with higher-than-average student count

Suggestion:

- Increase bus frequency
- Add additional pickup points

UNDERUTILIZED STOPS

Stops with very low student count

Suggestion:

- Merge with nearby stops
- Remove unnecessary stops

BALANCED STOPS

Stops with average student count

Suggestion:

- Maintain current operations

Strategic Improvements

- Optimize routes by focusing on **high-demand stops**
- Reduce operational cost by removing **low-demand stops**
- Improve scheduling based on **student density**
- Use data-driven thresholds instead of fixed assumptions

7. CONCLUSION

The analysis demonstrates that stop-level evaluation provides deeper insights compared to route-level analysis. By identifying overloaded and underutilized stops, the college can optimize its bus routes, improve efficiency, and enhance student convenience. Data-driven decision-making plays a crucial role in improving transportation systems.

